VIT BHOPAL · EPICS 2026–27

Metabolic City AI — Final Progress Report

Project: Urban Risk Management and Intelligence Platform

Institution: VIT Bhopal

Reporting period: 10 September 2026 — 10 March 2027

Report prepared: 11 November 2026

Final review: 10 March 2027

Status: Working end-to-end prototype with the MERN platform and Python intelligence capability set implemented.

Executive summary

Metabolic City AI is a working urban-risk platform for Bhopal, Indore and Sehore. It turns weather, road, terrain, traffic and citizen signals into a shared H3 cell view, then connects each cell to alerts, incident decisions, field work and an audit trail.

The project is deliberately split into two runtimes:

- **Node.js + Express + MongoDB** owns authentication, ingestion, scoring, alerts, incidents, field operations, administration and audit.
- **Python + FastAPI** owns cell memory, multi-resolution H3, graph analytics, hybrid retrieval and the LangGraph advisory workflow.
- **React + Vite** provides the operator, field, administrator and public reporting surfaces.

The implementation is evidence-first. Measured, interpolated, modelled, simulated and unavailable inputs are kept distinct. A missing source lowers confidence; it is not replaced with a believable zero. Playbooks are project-authored, agent runs are advisory, and forecasts are slope projections rather than trained predictions.

The current local baseline contains **100 H3 cells, 5,211 OSM road segments, 260 telemetry events, 150 alerts, 13 workflow/test incidents, 23 knowledge documents, 100 cell memories and 5 agent runs**. The repository has **95 commits**, approximately **33,362 lines** across the core source trees, and **25 automated test files**.

Delivery timeline

Period	Focus	Outcome
10–30 Sep 2026	Foundation	Authentication, MongoDB, telemetry engine, initial deployment and first product surfaces
1–31 Oct 2026	Operational platform	Modular React console, incident workflow, citizen reporting, analytics, TomTom traffic and request budgeting
1–14 Nov 2026	Evidence and intelligence baseline	Live alerts, forecasting, flood screening, incident lifecycle, realtime sync, data quality, documentation and decks
15–24 Nov 2026	Python review window 01	Cell memory and multi-resolution H3
25 Nov–4 Dec 2026	Python review window 02	H3 graph analytics and cascade context
5–14 Dec 2026	Python review window 03	Hybrid retrieval and Cloudflare Workers AI
15–24 Dec 2026	Python review window 04	LangGraph advisory workflow
25 Dec 2026–3 Jan 2027	Python review window 05	MongoDB change-stream realtime
4–13 Jan 2027	Python review window 06	Data quality and provenance
14–23 Jan 2027	Python review window 07	API contract and boundary review

Period	Focus	Outcome
24 Jan–2 Feb 2027	Python review window 08	Multi-resolution and graph performance review
3–12 Feb 2027	Python review window 09	Forecast audit and evidence review
13–22 Feb 2027	Python review window 10	Failure-mode and security review
23 Feb–4 Mar 2027	Python review window 11	End-to-end demonstration review
5–10 Mar 2027	Final review window	Freeze, final report and sign-off

The dated Python rows are the formal **10-day review and sign-off windows** for the final submission. The capabilities themselves are present in the repository; the dates are the review calendar, not a claim that each line of code was first written on that date.

Completed work: 10 September — 14 November 2026

10–30 September — Platform foundation

- Established the repository and initial authentication flow.
- Connected MongoDB Atlas with retry and timeout handling.
- Implemented JWT access and rotating refresh tokens.
- Added operator, field-crew and administrator roles.
- Built the first telemetry ingestion and H3 risk engine.
- Added the initial deployment configuration and municipal product surface.
- Completed the first set of operator, field and analytics pages.

Commit evidence: initial setup, MongoDB authentication, real-time telemetry engine, production deployment phases, municipal operations product surface and remaining pages (1, 10, 26, 27 and 29 September).

1–31 October — Operational platform

- Replaced the single-file frontend with modular React components.
- Completed the operator console, map, cell detail and action drawer.
- Implemented alert acknowledgement, incident approval, override, dispatch and resolution.
- Added field-crew task handling, GPS status, evidence upload and offline outbox behaviour.
- Added the citizen QR reporting path.
- Built the analytics dashboard, filters, incident views and CSV export.
- Integrated TomTom flow and incident data.
- Added monthly request budgeting and guarded provider calls.
- Added Redis-ready Socket.IO fan-out and SSE cell updates.
- Prepared the Render and Vercel deployment configuration.

Commit evidence: UI transformation, frontend refactor, incident workflow, citizen reporting, analytics, TomTom integration, request-budget controls and deployment preparation (1, 2, 21 and 22 October).

1–14 November — Evidence, realtime and intelligence baseline

- Made the alert and telemetry pipeline operate as a realtime system.
- Added the three-hour forecast lens and forecast snapshots.
- Added a screening flood forecast with an honest accuracy baseline.
- Surfaced live incidents as alerts instead of hiding them inside averages.
- Fixed traffic ingestion so road speed, incidents and mobility reach the console.
- Added citizen and field report verification states.
- Added reporter evidence and timestamps to report alerts.
- Added the incident lifecycle view to the operator cell detail.
- Implemented Cloudflare Workers AI embeddings and brief generation, with deterministic local fallbacks.
- Added MongoDB change-stream synchronisation for cell memory and incident retrieval.
- Added the live data-quality and provenance endpoint.
- Added the Intelligence page and data-quality panel.
- Updated the README, technical documentation and all three presentation decks.

Commit evidence: traffic repair, realtime pipeline, forecast lens, flood screening, report verification, incident lifecycle and operator fixes (4, 9 and 10 November).

Current verified baseline

Area	Current value
Spatial cells	100 — 43 Bhopal, 32 Indore, 25 Sehore
Road network	5,211 OSM road segments
Terrain	100 terrain cells
Telemetry	260 events — 100 weather, 100 terrain/GIS, 60 TomTom traffic
Alerts	150 — 28 open, 96 resolved, 26 expired
Incidents	13 workflow/test records — 5 approved, 4 escalated, 2 detected, 2 resolved
Knowledge base	23 documents — 13 historical incident summaries and 10 project-authored playbooks
Cell memory	100 materialised memory documents
Agent runs	5 advisory, dry-run traces
Forecast snapshots	5,124
Audit records	1,053
Automated tests	25 project test files — 13 backend, 12 frontend
Repository history	95 commits

Area	Current value
Core source size	approximately 33,362 lines across backend, frontend and Python service

Data-quality snapshot

Check	Current value
Synthetic telemetry	0
Simulated telemetry	0
Freshness	fresh
Weather interpolation	inverse-distance weighting; 100 of 100 weather events interpolated
Average nearest weather station	2.132 km
GTFS-Realtime rows	0; adapter dormant
Synthetic/simulated flags exposed in UI	yes

Weather at a cell centroid is modelled from provider observations. It is not a street-level gauge. The API returns `exact: false` and the interpolation distance alongside the value.

Python intelligence feature record: 15 November 2026 — 10 March 2027

The Python service is an internal FastAPI process. Node authenticates and proxies every call; the browser never receives a Python token or database credential. The service reads the shared MongoDB database and cannot dispatch a crew, send an SMS, change incident state or write operational records.

All capabilities below are implemented in the repository and are listed against the 10-day review window in which they are demonstrated and signed off.

15–24 November — Cell memory and multi-resolution H3

Implemented capability

- Per-cell risk, incident, resource and neighbour history.
- Hourly summaries, recurrence, volatility, seasonality and a cell personality profile.
- Next-hour and three-hour slope projections.
- Resolution-8 street cells, resolution-6 neighbourhood rollups and resolution-4 city rollups.
- Contribution shares that explain which child cells concentrate a parent's risk.

Evidence: `/v1/cells/{h3Index}/memory`, `/v1/cells/{h3Index}/multi-resolution`, 100 stored memory documents.

25 November–4 December — Graph analytics and cascade context

Implemented capability

- H3 adjacency graph built from stored cells.
- Degree, betweenness and closeness centrality.
- Critical-hub and higher-risk-neighbour detection.
- Weighted least-cost response routes.
- Cut-vertex detection for contingency review.
- Cascade direction recorded as context, not as an automatic decision.

Evidence: `/v1/cells/{h3Index}/graph?rings=2`, NetworkX graph construction, 100-node local graph.

5–14 December — Hybrid retrieval and Cloudflare Workers AI

Implemented capability

- Hybrid ranking: 0.50 semantic, 0.25 spatial, 0.15 temporal and 0.10 outcome.
- Cloudflare Workers AI embeddings using `@cf/baai/bge-small-en-v1.5` at 384 dimensions.
- Cloudflare brief generation using `@cf/meta/llama-3.3-70b-instruct-fp8-fast`.
- Deterministic local hashed vectors and template briefs as the no-credential fallback.
- Vectors stored on `knowledge_documents` in MongoDB; no separate vector database.
- Project-authored playbooks marked `isOfficial: false`.

Evidence: `/v1/cells/{h3Index}/retrieve`, `python-service/app/rag_service.py`, 23-document corpus, knowledge rebuild endpoint.

15–24 December — LangGraph advisory workflow

Implemented capability

- Eight nodes: detect, risk assessment, context retrieval, resource allocation, communication, execution, escalation and learning.
- Evidence assembly from cell state, alerts, forecasts and retrieved documents.
- Recommendation and dry-run notification plan.
- Hard stops: no dispatch, no SMS, no incident-status mutation and no online retraining.
- Outcome samples stored for later offline learning.

Evidence: `python-service/app/agents.py`, `agent_runs` collection, five recorded dry-run traces.

25 December–3 January — MongoDB change-stream realtime

Implemented capability

- Change streams keep cell memory and incident retrieval documents current as the Node pipeline writes.
- Standalone MongoDB automatically falls back to short polling.
- Sync mode and connection state exposed at `/v1/realtime/status`.
- Realtime updates remain derived state; the operational record stays authoritative.

Evidence: `python-service/app/realtime.py`, `/v1/realtime/status`, change-stream and polling paths.

4–13 January — Data quality and provenance

Implemented capability

- `GET /v1/data-quality` reports source counts, freshness, synthetic/simulated flags, interpolation method and nearest-station distance.
- Node proxy at `GET /api/v1/intelligence/data-quality`.
- Per-cell `dataQuality` block in the intelligence response.
- Intelligence page panel with 30-second refresh for the overall snapshot and selected cell.
- Explicit computed-score, interpolation and weather-age labels.
- Current local result: 260 telemetry events, 0 synthetic, 0 simulated, weather 100/100 interpolated, average nearest station 2.132 km.

Evidence: `python-service/app/data_quality.py`, `backend/src/routes/intelligence.js`, `frontend/src/pages/Intelligence.jsx`.

14–23 January — API contract and boundary review

Implemented contract

- Node authenticates and proxies every intelligence call.
- Python routes are internal and return advisory output only.
- Every response identifies its evidence, source and confidence.
- The browser never receives a database credential or Python service token.
- Missing data is represented as missing, not zero.
- Playbooks and agent output are labelled as advisory.

Evidence: `/api/v1/intelligence/*` proxy routes, FastAPI route table, README boundary section.

24 January–2 February — Multi-resolution and graph performance review

Implemented review

- Cell memory and graph endpoints operate over the 100-cell local grid.
- Multi-resolution rollups use real H3 parent membership rather than string truncation.
- Graph construction is limited to stored cells; missing cells are not invented.
- Local request timings are sufficient for interactive use.
- No production throughput or metropolitan-scale claim is made.

Evidence: local endpoint checks, 100-cell graph, 5,211 road segments and 100 terrain cells.

3–12 February — Forecast audit and evidence review

Implemented audit

- 5,124 forecast snapshots stored with horizon, predicted risk, inputs and issue time.
- Forecasts use slope extrapolation: next hour = current + slope × 0.6; next three hours = current + slope × 1.8.
- The forecast is explicitly not a trained model.
- Stored snapshots allow later comparison against observations.

- No accuracy claim is made until issued forecasts are compared with observed outcomes.

Evidence: `forecastsnapshots`, forecast-accuracy route and forecast audit tests.

13–22 February — Failure-mode and security review

Implemented safeguards

- Strict CORS allow-list and Helmet headers.
- JWT access tokens, rotating refresh sessions and revocable logout.
- Role checks on protected routes.
- Idempotency keys and telemetry dedupe hashes.
- Provider circuit breakers and a dead-letter queue.
- Rate limits, 100 kB JSON body cap and graceful shutdown.
- Explicit degraded states when weather, traffic, transit or Python is unavailable.
- No synthetic fallback is presented as measured data.

Evidence: middleware, ingestion services, provider status routes and health endpoints.

23 February–4 March — End-to-end demonstration review

Implemented flow

1. Telemetry enters through the Node ingestion path.
2. The H3 risk engine builds the cell projection.
3. Alerts and incidents enter the operator workflow.
4. The operator locks, decides, dispatches and resolves.
5. The field crew acknowledges, arrives and closes the task.
6. Python returns memory, graph, retrieval and advisory agent context.
7. The audit trail records the decision path.

Evidence: operator console, field crew console, incident state machine, intelligence panel and audit log.

5–10 March — Final review and freeze

Final deliverables

- Working local stack: Node on 5000, Python on 8001 and Vite on 5173.
- 100-cell operational dataset with live weather and TomTom traffic.
- Data-quality and provenance panel.
- Python intelligence service with memory, multi-resolution, graph, RAG and LangGraph.
- README, technical theory deck, pitch deck and updated report.
- Explicit limitations, non-claims and known gaps.

10-day review and sign-off calendar

Formal review every 10 days from 15 November 2026 to 10 March 2027. Informal weekly stand-ups continue between formal reviews.

#	Review date	Review focus	Evidence reviewed	Student sign	Guide sign
01	15 Nov 2026	Python service boundary and baseline	FastAPI health, Node proxy, Mongo connection	_____	_____
02	25 Nov 2026	Cell memory and multi-resolution	Memory and parent/child endpoints	_____	_____
03	5 Dec 2026	Graph analytics	Centrality, critical hub and routes	_____	_____
04	15 Dec 2026	Hybrid RAG and Cloudflare	Retrieval ranking, embeddings, rebuild	_____	_____
05	25 Dec 2026	LangGraph advisory workflow	Eight-node trace and dry-run boundary	_____	_____
06	4 Jan 2027	Realtime synchronisation	Change streams, polling fallback, status	_____	_____
07	14 Jan 2027	Data quality and provenance	`/v1/data-quality` and UI panel	_____	_____
08	24 Jan 2027	API contract and access boundary	Proxy routes and response contract	_____	_____
09	3 Feb 2027	Resolution and graph performance	100-cell rollups and graph timings	_____	_____
10	13 Feb 2027	Forecast audit	Snapshot coverage and non-claims	_____	_____
11	23 Feb 2027	Failure modes and security	Provider failure, auth and degraded states	_____	_____
12	5 Mar 2027	End-to-end demonstration	Operator-to-field-to-audit flow	_____	_____
13	10 Mar 2027	Final freeze and report	Repository, decks, report and demo	_____	_____

Known limitations and controls

Limitation	Control
No live Bhopal municipal feed	State the boundary explicitly; do not claim a municipal deployment
GTFS-Realtime has zero rows	Keep the adapter dormant until a compatible feed is configured
13 incidents are workflow/test records	Label them as test records, not municipal history
Weather is interpolated	Show interpolation method, distance and `exact: false`
Vulnerability is a terrain proxy	Do not present it as household or social vulnerability
Forecasts are slope projections	Store snapshots and make no accuracy claim without comparison
Agent output is advisory	No dispatch, SMS, status change or online retraining
Playbooks are project-authored	Return and display `isOfficial: false`
Python service has no automated test suite yet	Add contract and failure-mode tests before any production claim
No production load test or SLA	Report local verification only
No public deployment	Keep the local demo reproducible and document startup

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Definition of done

The project is ready for final review when:

- The Node, Python and frontend services start together.
- The data-quality panel reports zero synthetic and zero simulated rows.
- The operator can move a workflow/test incident from detection to resolution.
- The field crew can receive, accept, arrive and resolve assigned work.
- Python returns memory, multi-resolution, graph, retrieval and advisory output.
- The browser never calls Python directly.
- Missing data remains missing and is reflected in confidence.
- The README and all three decks match the running implementation.
- No claim is made about municipal adoption, production SLA or measured response-time improvement.
- The final report and sign-off calendar are complete.

Appendix: commit history reviewed

The report was prepared after reviewing the complete local Git history: **95 commits** across September, October and November 2026.

Date	Representative milestone
1 Sep 2026	Initial setup, login and sign-up logic
10 Sep 2026	MongoDB authentication flow
26 Sep 2026	Real-time telemetry engine
27 Sep 2026	Production deployment phases and municipal product surface
29 Sep 2026	Remaining operator, field and analytics pages
1 Oct 2026	Dark UI transformation and initial sample data
2 Oct 2026	Incident workflow, frontend componentisation and login/homepage fixes
21 Oct 2026	TomTom traffic, monthly request budgets and free-tier controls
22 Oct 2026	Deployment preparation and README update
4 Nov 2026	Traffic data repair and documentation update
9 Nov 2026	Realtime alert pipeline, three-hour forecast and flood screening
10 Nov 2026	Incident lifecycle, report verification and operator console fixes

Repository: `Puneet04-tech/Metabolic_City` **Branch:** `main` **Current local baseline verified:** 11 November 2026
Final review: 10 March 2027